JEWERS
The Mark of Excellence & Innovation

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BUDGET QUOTATION

PROPOSED NEW AIRCRAFT HANGAR FOR INDIAN DEFENCE PROJECT

ESAVIAN TYPE 126 SLIDING HANGAR DOORS BY JEWERS

DESIGN, MANUFACTURE, EX-WORKS

ONE set of **Esavian** TYPE 126 power operated, bottom rolling Steel Sliding Doors and Top and Bottom Tracks to suit an overall clear opening approximately **51,000 mm (W)** x **14,000 mm (H)**.

The set of hangar doors is formed in SIX equal leaves, each approximately 8,585 mm (W) which slide on THREE parallel tracks spaced at approximately 715 mm centres.

THREE leaves slide to each side clear of the opening into enclosed parking areas to each side of the opening. The preferred door configuration is such that the two centre (Master) leaves #3 and #4 are on the innermost track and the trailing leaves #1 and #6 are on the outermost track. The overall length of track is approximately 68.5 metres.

The overall thickness of each leaf is approximately 605mm. In addition, the seal gap between leaves on adjacent tracks is approximately 110 mm. The overall depth of the door pack is therefore approximately 2,145 mm (i.e. 3 x 605 + 3 x 110). The clear bunching space required inside each door pocket is 9,100 mm wide x 2,700 mm deep.

The overall weight of the full set of sliding doors inclusive of electrical, mechanical equipment, top, bottom tracks and cladding will be approximately 65,000 kg i.e. up to 9.5 tonnes per 8.5 m (W) leaf.

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DOOR CONSTRUCTION

The door is to be designed to withstand local wind speeds which are factored in accordance with ASCE 7-05 “Minimum Design Loads for Buildings and Other Structures”.

The following input data are used:

- Method 2 from within ASCE 7-05.
- Basic Wind Speed = 300 kph for a 3 sec gust, (equivalent to 83 metres / sec).
- Importance Factor = 1.0.
- Exposure Category = “C”.

All main structural steel elements of the door leaves are designed in accordance with BS EN 1993-1-1 Eurocode 3: Design of Steel Structures Part 1-1: General Rules and rules for buildings and the UK National Annex.

Maximum deflection of any main structural member is L/140.

Permanent and Imposed Loads

The doors to be designed to withstand permanent and imposed loads according to BS EN 1991-1-1

Structural Design

All main structural steel elements of the door leaves are designed in accordance with AISC 360-10 Design of Steel Structures.

Structural Joints

Structural joints used within the door structural to be design according to AISC 360-10. All such joints to be bolted together using ordinary (non-preloaded) assemblies.

Steel Materials

Our door leaf design is based upon the use of steel shapes, principally square or rectangular hollow sections in accordance with BS EN 10210 Part 2: 1997 to minimum Grade S355 in accordance with BS EN 10025: 2004 and BS EN 10210 Part 1: 1994.

Execution Class

As defined by the afore-mentioned general design criteria, fabrication and erection of all structural steel relating to the door structures to be carried out according to Eurocodes BS EN 1090-1 and BS EN 1090-2.

- Execution to be carried out to Execution Class EXC2
- Setting out, manufacturing and erection tolerance to Class 1
- Inspection levels to level IL2.

PROTECTIVE COATING / PAINTING

Prior to fabrication, all main, hot rolled, structural sections are treated as follows:

1. Shotblast to SA2½
2. Apply 25 µ DFT twin-pack zinc-rich pre-fabrication primer

After fabrication all main structural sections to be further treated as follows:

3. Apply 100 µ DFT twin-pack, epoxy zinc phosphate high build primer.
   (Leigh's Epigrip C400 or equivalent)

Further painting is not included at this stage although superior alternatives are available at extra cost.

Bottom tracks are hot dip galvanised to a minimum 85µm.

CLADDING SYSTEM

At this stage each door leaf structure is designed to accept the cladding materials as per the following:

- Main vertically-laid profiled steel cladding system.
- Internal low level protective panelling.

Further details of each of these materials are as follows:

1. Vertically-Laid Profiled Aluminium Cladding System. - Excluded

Each door leaf to be clad, full width and full height on the external face only with a single skin, profiled aluminium cladding system.

Profile is 0.70 mm thick aluminium, finished in HPS-200, standard colour. The face is 32 mm deep profiled.

All flashings are produced from matching 0.70 mm thick HPS-200-coated material. Fixings comprise stainless steel self-drilling screws, all as per SFS-Intec or other supplier's standard specification. Colour caps are not used.

2. Low Level protective panelling to mechanical equipment (included).

On the inside, certain areas of the door (such as drive compartments) will require access internally: these areas are fitted with access panels which are secured using lockable latches.

PERSONNEL DOORS

None fitted available on request.

WEATHERSEALS

All **Esavian** seals are designed to resist draught and dust.

All vertical joints between leaves on adjacent tracks and the trailing-end jamb are fitted with EPDM wiper seals set in aluminium extrusions. Suitable flashings are to be provided **by others** at the trailing-end jamb to which this seal may be secured using self-drilling, self-tapping screws. The vertical wiper seals mounted on the doors will be fitted using the same screws.

At the butt joint between the two leaves that meet at the centre-line of the hangar, a compressible EPDM "bulb" seal is fitted. This is retained in an aluminium extrusion.

At the bottom, each leaf is fitted externally with a nylon brush seal, set in aluminium extrusions.

At the top, leaves are fitted with extended flashings which project into the top track to minimise any airflow in the area. The provision of staggered soffit cladding directly in front of the hangar doors is to be **by others**.

All aluminium is supplied in the mill-finish condition and the EPDM is black in colour.

BOTTOM WHEELS AND BOTTOM TRACKS

Each 8.65 m (W) leaf is supported at the bottom on 2 No. heavy duty double-flanged steel track wheels mounted on tapered roller bearings. Wheel diameter is approximately 260 mm.

Wheels are manufactured from flame-hardened EN24T steel. Each wheel is mounted in a wheel box so that the complete wheel may be removed if maintenance is necessary. Grease nipples are provided for lubrication.

The bottom tracks comprise 3 No. parallel runs of S10 (or equivalent) flat bottom rail mounted on transverse bearer members at approximately 1,500 mm centres. These tracks extend only over the distance over which the leaves are free to move.

A continuous trench for the bottom tracks, approximately 2,500 mm wide x 160 ± **10 mm** deep, is required. Channels formed in the concrete **by others** provide clearance for the wheel flanges.

It is recommended that some form of drainage be also incorporated into the bottom track foundation so that water drains away from the area in front of the door rails into an external channel-type drain. Any such drainage is to be provided and cast into the concrete **by others**.

TOP TRACK SYSTEM AND TOP ROLLERS

Top tracks are included within our scope of supply. These include the following:

- Fully prepared sections of top guide track.
- Suitable packers and fixings to allow connections to the pre-drilled horizontal bearer sections (spaced at MAXIMUM 2.5 METRE CENTRES across the full width of the opening.

Top tracks comprise a high-yield, alloy steel "C"-shape The C-section is bolted direct to the underside of a series of transverse bearers provided which are to be provided **by others** at approximately 400 mm higher than the nominal clear opening height and at approximately 2.5 metre **maximum** centres or as otherwise agreed. It is assumed that the top track supporting steelwork will have been accurately pre-drilled **by others**, according to details agreed between the relevant parties.

The approximate weight of the top track system is 55 kg per linear metre (on average).

The whole of the top track system is hot dip galvanised in accordance with EN 10147 to a thickness of 85 µ DFT and left in that finish.

Each leaf is fitted at the top with 2 No. telescopic top roller units. The top rollers comprise a trolley head unit which is connected to a vertical plunger unit which is free to rise or fall vertically within low friction bearing guides mounted in a vertical tube within the door leaf. The trolley head unit runs within the "C"-section top track which supports its deadweight, and to which it transmits wind loads.

The top rollers and top tracks are able to accommodate live load deflections or deviations in the top tracks of up to L/360 (where "L" is the minimum roof span dimension) e.g. for a 51.0 m (W) span, allowable deflection = 51,000 ÷ 360 = 142 mm total or ± 71 mm.

BUFFERS

Heavy duty energy absorbing buffer stops are provided at the extremities of the bottom tracks to cushion and stop the doors coming off the end of their tracks in an emergency situation. Although designed to accommodate maximum impact loads, these buffers are not used in normal door operation.

DOOR OPERATION

For the purposes of power operation the THREE leaves filling each half of the opening are considered as a set of ONE Master leaf and TWO Slave leaves. Each Master leaf is fitted with two drive units which are each arranged to drive a bottom wheel.

The Slave leaves are connected to the Master leaf by means of a steel cable sheaving system so that all three leaves move simultaneously, but at differential speeds, and all reach their open or closed positions at the same time.

The sheaving system comprises a series of galvanised steel cables and pulleys which are mounted approximately 7,000 mm above the ground and link the leaves together.

The system is positive so that if the Master leaf is stopped, the Slave leaves will also stop and **not** overrun. Neither the cables nor the pulleys and mounting brackets are seen from outside the building. A suitable mounting point is required on each jamb stanchion and at the end of the door pocket to which this equipment may be anchored.

Each drive unit comprises a minimum 1.5 kW, 4 pole, squirrel cage motor with integral magnetic brake and a helical-worm reducing gearbox. The unit drives a bottom wheel through a single stage simplex roller chain. The brake motor unit is supplied by SEW Eurodrive.

Each drive unit is enclosed in a steel compartment complete with main internal access cover. The access covers are interlocked to prevent operation of the drive units whilst either of them is open.

The normal speed of door operation is 15 metres per minute from which speed the door leaf would be expected to stop within approximately 100 mm in dry rail conditions. Opening time per group of three leaves is therefore approximately 2 minutes. The drive units are capable of safely operating the door at all wind speeds up to 20 metres per second (approximately 40 knots).

In the event of power failure the brakes on the drive units may be disengaged and the door easily towed along using the towing pins provided.

ELECTRICAL CONTROL PANEL

A feature of **Esavian** door design is that principal electrical controls, limit switches and door sensors are kept at low level for ease of maintenance.

A separate enclosed compartment is formed at the side of one drive unit which houses the main electrical control panel for the Master leaf. The enclosure is steel to IEC classification IP55.

Equipment includes secondary low voltage control circuit, alarm signals incorporating time delay, with terminal board to allow connection of all external devices (push buttons, alarms, limit switches etc.) to the control panel. In addition, the control panel is fitted with an inverter-controlled, variable speed drive system.

The variable speed drive enables the setting of different speeds for the door, but more importantly enables the rate of acceleration and braking of the door to be accurately set up. The torque in the drive unit during the critical starting and stopping phases is precisely controlled and is free of snatch.

Deceleration of the door is controlled by the inverter and its brake resistor, and only when the door has been brought to rest is the mechanical brake applied. Wear, and the maintenance cost of the drive unit is greatly reduced.

DOOR CONTROL SYSTEM

General Principles

The **Esavian** hangar door control system is designed to comply with the BS EN standard for power operated industrial doors (BS EN 12453: 2017, “Industrial, commercial and garage doors and gates – Safety in use of power operated doors – Requirements and Test Methods”). The principle requirements of this standard which are relevant to this installation are as follows:

- The person controlling the door shall be in full view of the door and in the vicinity of the door during the leaf movement.

- When a door leaf is within 500 mm of its fully closed position it must be capable of being stopped within just 50 mm (which compares with a stopping distance of 75 mm from full running speed).

The ways in which the standard **Esavian** hangar door control system is designed to comply with each of these requirements is as follows:

- Each Master leaf is fitted at the leading edge with a pendant push button station. The pendant is mounted to the leading edge of the leaf on a cord about 2 metres long that allows the operator to move from side to side to see where the door is being driven. The pendant is hung up on a hook when not in use.

The buttons themselves include an “OPEN” and “CLOSE” button and an “EMERGENCY STOP” button. Pushbuttons are of the dead-man's type with the exception of the "EMERGENCY STOP" button which is a "twist-to-reset" type. In addition a “Power ON” indicator lamp is fitted.

- A 2-stage limit switch assembly is fitted to each Master leaf. This allows a door firstly, to be *automatically* slowed from full speed to slow speed when the door is approximately 500 mm from its end position and secondly, to stop automatically in its closed position.

  Limit switches comprise automatic “proximity sensor” closure devices which are fitted in to the bottom of the Master leaf. These allow the operator to stop the group of leaves automatically and accurately in its closed position, where it can be guaranteed that the seals at the lap joints will be accurately made – every time.

  The proximity sensors are solid state devices with no moving parts. Hence they are highly reliable and in any case are very easy to gain access to and adjust if necessary.

Electrical Control Panel
A feature of **Esavian** door design is that principal electrical controls, limit switches and door sensors are kept at low level for ease of maintenance.

The main electrical control equipment for each Master leaf is mounted inside the leaf. The enclosure is steel to IEC classification IP55.

Equipment includes a 3-phase vector type variable speed drive with LCD display and brake resistor, PLC with fault tracing LED’s, miniature circuit breaker, and 24 Volt secondary transformer. The PLC provides all of the logic functions and terminal connections for the door components in a single standard multi-purpose unit. The PCB is highly reliable and versatile, and even enables additional devices to be added at a later date if required.

The variable speed drive enables the setting of different speeds for the door, but more importantly enables the rate of acceleration and braking of the door to be accurately set up. The torque in the drive unit during the critical starting and stopping phases is precisely controlled and is free of snatch. Current consumption during starting and stopping is much reduced from conventional Direct-on-Line Contactors.

Deceleration of the door is controlled by the inverter and its brake resistor, and only when the door has been brought to rest is the mechanical brake applied. Wear, and the maintenance cost of the drive unit are greatly reduced.

SAFETY AND WARNING DEVICES

Each Master leaf is fitted on the inside face with an audible and a visual alarm, both of which operate at all times the group of leaves is in motion under power, and also for a period of 5 seconds before the leaf starts to move.

The audible alarm is a high output alarm emitting 106 dB at 1 metre range. The tone may be adjusted. The visual alarm is a single strike xenon beacon fitted with an orange lens (or other preferred colour).

Interlock switches are fitted to the access panels to the drive units and electrical control panels, to prevent the leaf on which they are fitted being used while these are open.

POWER SUPPLY

The incoming power supply to the hangar doors is 400 volts 3 phase and earth, 50 Hz or as agreed.

Power is transmitted to each of the Master Leaves by means of an of enclosed duct type overhead conductor system mounted inboard of the door head track area. Each Master leaf draws a collector trolley through the duct.

It is anticipated that the system will be end-fed i.e. a 32 Amp, isolatable power supply will need to be made available (**by others**) towards **both sides** of the opening at high level.

Hard wiring within the Master Leaves is to be plastic-sheathed steel wire-armoured (SWA) cable without conduit, secured with cable clips (included). Minimum core size is 1.5 mm^2. This wiring is supplied by ourselves.

The scope of supply of our wiring materials will terminate at one end of each high level conductor system. All wiring up to the junction boxes at each end of the system is to be **by others**. It is recommended that isolators are provided **by others** at low level on each jamb.

SHIPPING SPECIFICATION – by OTHERS

Overall weight of materials within our basic offer is approximately **60 tonnes**.

Materials can be transported by sea in up to 4 No. 40' Open Top containers and 1 No. out of Gauge Container.

The Tariff Number for goods within our supply which we have used for export projects in the past is **730830 00 0 (Structures: - Doors and Windows)**.

INSTALLATION

At this stage, we do not include for installation, however we do provide a comprehensive installation manual.

SUPERVISION

We offer an **extra** for the services of an experienced engineer to supervise the installation. We would estimate that a maximum of 3 working visits, each of one week's duration would be required. We believe our engineer might be present at the following stages:

i)	Laying bottom tracks and assembling first leaves	1 week.
ii)	Lifting first leaves	1 week.
iii)	Final stages of installation including commissioning	1 week.

The price we offer for the engineer's services is in addition to the basic contract value and **excludes** his airfares and all accommodation, subsistence and travel expenses whilst he is away from UK.

TRAINING

It would be very advantageous if you are able to send one or more of your installation personnel over to England prior to erection commencing to be trained by us and see some of our existing installations.

Typically such a visit would last one week and could reduce the likelihood of further visits by our engineer to SITE.

Although training is offered free of charge, airfares, local hotel and subsistence expenses are **not** included. Local transport whilst your personnel are at our works will be provided.

OPERATING AND MAINTENANCE MANUALS

We include for the provision of 2 hard copies of the "Operation and Maintenance Manual", written in the English language and a digital CD version.

EXCLUSIONS

1. All cladding sheets and flashings required for the doors are **excluded** from our supply.

2. Any other metal wall panels and associated flashings, fixings and the like materials required for sheeting the doors are **excluded** from our supply.

3. All steelwork for connecting our top tracks to, are **excluded** from our supply.

4. Provision of suitable mounting points for various items e.g. jamb seals, sheaving system connections points etc. are **excluded** from our supply.

5. Additional painting, other than specified above, is **excluded** from our supply.

6. Delivery beyond CFR Port of Mumbai any demurrage costs, import duties and taxes are **excluded** from our supply.

7. Additional design modifications and materials are **excluded** from this offer.

8. Material submittals other than weatherseals samples, paint specification and electrical submittal paper copies only.

9. Any costs associated with a third party engineer checking our calculations.

10. Delivery of materials

PRICE SUMMARY

For the Sum of **£292,190.00** net.

(TWO HUNDRED AND NINETY TWO THOUSAND ONE HUNDRED AND NINETY POUNDS STERLING)

Schedule of Extras

Extra for Site Supervisor – **£3,950.00** per week net.

(THREE THOUSAND NINE HUNDRED AND FIFTY POUNDS STERLING)

All prices are quoted in GBP.

GENERAL CONDITIONS

Delivery: General arrangement drawings to be submitted 3 to 4 weeks from order and agreement of all technical details enabling us to produce full drawings.

Shipping of materials can then commence a further 14 weeks from full approval of GA drawings with the final (4th) container being dispatched at 18 weeks. All delivery periods are based on Ex Works. Shipment to Port of Mumbai ex-UK is approximately another 5 weeks.

Payment: Full payment to be secured by an irrevocable letter of credit, confirmed and payable in sterling by a bank in the United Kingdom. This will be opened for 100% of the contract value in our favour at your expense and payable upon presentation of the relevant shipping documents. The credit is to be valid for the full delivery / shipment period described in this document plus a further 21 days thereafter for registration of shipping documents. Part shipments to be allowed. Modifications to the L.C. to be at the buyer's expense.

Import Duties: The above prices **exclude** any shipping, import duty, custom or freight forwarding / clearance charges which may be payable on importation of materials into India or other transit / destination country.

Other Terms: All other terms to be as indicated on the front sheet of this quotation.

Price: Our prices are current and subject to fluctuation.

For Jewers Doors Limited

J.Jewers

Jonathan Jewers